

NeuroAdapt: The Autonomous Neuro-Diverse Learning Ecosystem

Abstract

Approximately 15 to 20 percent of the global population is neurodivergent, representing over one billion people whose cognitive profiles diverge from neurotypical norms across conditions such as ADHD, dyslexia, dyscalculia, and autism spectrum conditions. The economic consequence of failing this population in education is measurable and severe: ADHD alone is associated with a \$122.8 billion annual societal excess cost in the United States, with unemployment and workplace productivity loss accounting for nearly \$96 billion of that figure, a burden that traces directly to lower educational attainment during the formative years. Despite this, the OECD notes that while AI and advanced technologies hold clear promise for neurodivergent learners, significant structural and design barriers continue to prevent that potential from being realised in practice. Most existing EdTech platforms replicate the failures of the traditional classroom in digital form: they change the medium but leave the pedagogy unchanged.

NeuroAdapt is proposed as the first system to unify three advanced technology paradigms: agentic AI, hybrid quantum-classical computing, and immersive generative content synthesis into a single, diagnosis-agnostic adaptive learning ecosystem. The global neurodivergent-focused EdTech market is valued at \$2.4 billion in 2025 and projected to reach \$6.4 billion by 2032 at a CAGR of 15.1%, with personalised adaptive learning platforms identified as the largest and fastest-growing segment, and Asia Pacific, including India, projected as the highest-growth region. NeuroAdapt is positioned precisely at this intersection: a technically novel, commercially viable, and socially urgent system that treats neurodivergent cognitive profiles not as exceptions to be accommodated, but as the primary design case for a more intelligent and equitable education infrastructure.

Introduction

Digital education currently operates under a 'one-size-fits-all' model that creates significant barriers for neurodivergent learners. For students with ADHD, dyslexia, or dyscalculia, this lack of flexibility often leads to cognitive overload and executive function paralysis. It is not that the student lacks potential, but rather a design flaw in modern platforms that shuts out those who process information differently. NeuroAdapt addresses this integral failure by replacing static instruction with a responsive and empathetic dialogue. Instead of a one-way broadcast of information, the platform functions as an interactive ecosystem that adjusts to the student in real time. By prioritising individual minds over standardised delivery, our design ensures that the technology adapts to the learner, rather than forcing the learner to struggle with the technology.

Problem Statement

In India, the financial cost of supporting a neurodivergent learner falls almost entirely on the family. Specialised schooling and intervention centres cost between Rs. 2.5L and Rs. 8L per year, constructing a socio-economic barrier that excludes the majority before they even enter a digital classroom. Beyond cost, the conventional model causes a developmental cost: students experience social isolation, persistent burnout, and executive function paralysis, all of which accelerate dropout and erode self-esteem. The

system ties access to support to a formal clinical diagnosis, which is expensive and exclusionary in its own right. NeuroAdapt is a diagnosis-agnostic ecosystem that places the responsibility of adaptation on the technology, not the student, and treats each neurological profile as an asset rather than a liability.

Literature Review

Research consistently establishes that neurodivergent students face structural, not intellectual, barriers in education. College students with ADHD maintain significantly lower GPAs and are less likely to persist through eight semesters compared to neurotypical peers, with inattention symptoms identified as the primary driver of academic underachievement rather than cognitive ability. Neurodivergent students as a group are 26% less likely to complete courses than their peers, and higher education institutions have largely failed to close this gap through existing accommodation frameworks.

On the technology side, reinforcement learning has demonstrated meaningful improvements in personalised tutoring. RL-based intelligent tutoring systems outperform non-adaptive rule-based systems in test accuracy and learner engagement, with one study reporting a 30% reduction in hint requests and higher post-test scores among students taught by RL-driven systems compared to static counterparts. For the quantum component, recent work on hybrid quantum-classical architectures confirms that VQCs consistently accelerate training convergence compared to purely classical models, with the primary advantage lying in faster loss reduction during early training epochs rather than in final accuracy. This directly supports NeuroAdapt's use case, where rapid per-learner adaptation within the first few sessions matters more than marginal gains at scale.

Objectives

NeuroAdapt is built around three core goals:

- Close the achievement gap by enabling real-time, personalised learning environments that target the approximately 20% of undergraduates who are neurodivergent, with a measurable aim of increasing learner engagement by 40% and significantly reducing dropout rates.
- Shift from accessibility to stability by moving beyond surface-level accommodations like closed captions toward a "Stability-Focused" model, where the system actively manages a student's cognitive load and maintains it at optimal levels throughout each session.
- Serve the undiagnosed majority by training on behavioral signals rather than clinical labels, ensuring that students who have never received a formal diagnosis, and who currently fall through every institutional support structure, receive the same quality of adaptation as those who have.

Methodology

NeuroAdapt operates through four coupled layers that together convert a passive learning session into a continuously self-correcting experience.

The Observer: a privacy-first, client-side telemetry layer that captures five behavioral signals every 30 seconds without any camera or audio input, guarding user privacy as well. These are Semantic Dwell

Ratio (time-on-screen relative to word count, detecting reading walls or impulsive skimming), Interaction Jitter (mouse velocity variance, signalling anxiety or flow), Focus Persistence (tab-switch frequency, measuring dopamine-seeking behaviour), Stall Duration (time since last interaction, distinguishing active thinking from executive function paralysis), and Preference Delta (the learner's explicit end-of-lesson format choice, serving as ground truth). Together these form a five-dimensional State Vector, i.e. a real-time fingerprint of the student's cognitive state.

The Orchestrator: ingests this vector and selects one of six interventions: hold course, issue a soft nudge, simplify the text, switch to video, inject a gamified task, or enforce a sensory break. This decision is made by a Hybrid Quantum-Classical Policy Network: a Double Deep Q-Network whose inner hidden layer is replaced by a Variational Quantum Circuit (VQC) built in PennyLane. The five behavioral features are encoded as qubit rotation angles, entangled via CNOT gates to capture correlated signals (such as the ADHD hyperfocus state, where high focus and high jitter co-exist), and measured as Z-basis expectation values. The VQC explores a higher-dimensional Hilbert space with fewer parameters than a classical network, enabling faster convergence on a learner's unique cognitive profile, and an overall better performance.

The Generative Synthesis Engine: executes the Orchestrator's content interventions asynchronously, with a predictive pre-fetching mechanism that begins generating the next format variant before the learner even finishes the current slide. Text is restructured by an LLM for the appropriate reading level; visuals are rendered via Stable Diffusion or HeyGen avatar video; audio is delivered through ElevenLabs TTS on a calm and neutral preset calibrated for autism or high-anxiety profiles.

The Feedback Loop: closes the reinforcement cycle at two points: an end-of-lesson format choice modal feeds explicit preference back into the policy weights, while a student-accessible Energy Bar acts as a manual break override. This registers as a high-magnitude negative reward that teaches the Orchestrator to pause earlier in future sessions. The system develops a richer model of the learner with every session, compounding in accuracy over time.

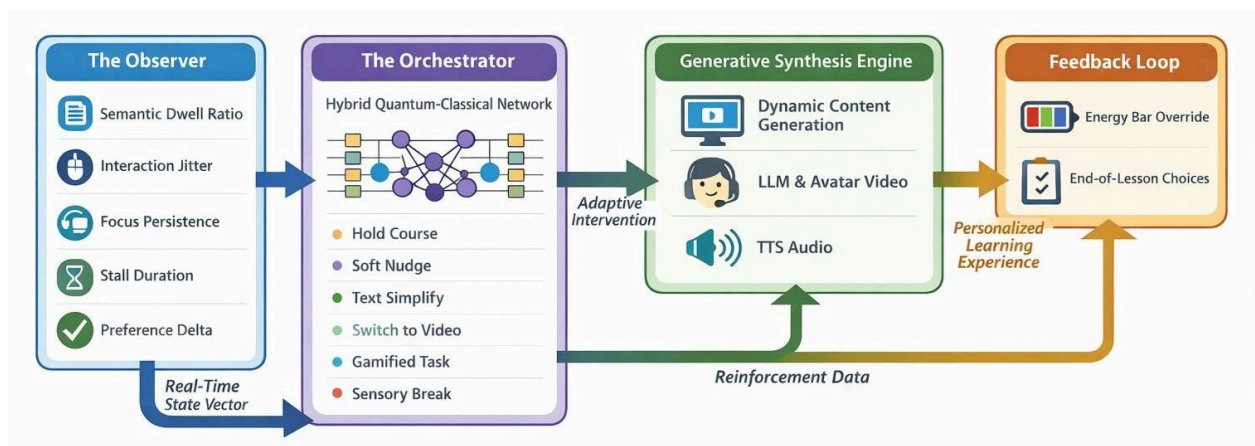


Figure 1: NeuroAdapt Methodology Flowchart

Expected Outcomes

NeuroAdapt targets four concrete, measurable outcomes:

- 25% improvement in module completion rates for neurodivergent learners, driven by real-time format adaptation that prevents the cognitive overload responsible for mid-session drop-off, consistent with comparable agentic AI deployments that have reduced STEM dropout rates by up to 41%.
- 10-15% gain in assessment performance, achieved not by lowering the bar but by delivering content at the right moment and in the right format, ensuring that difficulty is the only challenge the student faces.
- Faster policy convergence with the quantum-classical hybrid, where the VQC's entanglement of correlated behavioural signals is expected to produce an accurate per-learner model in fewer sessions than a classical baseline, with the Preference Delta trending measurably toward zero across a pilot cohort.
- Inclusive, diagnosis-agnostic support for neurodivergent students who remain undiagnosed and ineligible for formal accommodations, as NeuroAdapt responds to behavioural signals rather than clinical labels.

Collectively, these outcomes advance a single defining proposition: that a system intelligent enough to understand how a student is thinking, not just what they are answering, is not a luxury. It is the minimum standard a modern educational platform should meet.

Cost Optimisation and ROI

NeuroAdapt is built to be affordable at every layer. The Observer runs entirely in the browser at zero server cost, and the VQC is intentionally shallow, keeping inference overhead negligible. The only significant expense is generative API calls, which are gated behind a 60% confidence threshold so content is only produced when the system genuinely needs it. Open-source substitutes such as Llama-3 and Coqui TTS can replace paid APIs entirely for budget-constrained deployments. The ROI case is straightforward: preventing a single course failure for a neurodivergent student saves more in re-enrolment costs than the platform's full annual operating expense for that cohort.

Security and Scalability

All five behavioral signals are computed locally in the browser and only the normalized State Vector is transmitted to the server, meaning no raw interaction data or personally identifiable information ever leaves the client. This architecture satisfies the data minimization requirements of India's DPDP Act (2023) by design rather than by policy. On the scalability side, the inference layer is stateless and scales horizontally without architectural changes, and the NoSQL session store shards naturally by user ID. The VQC runs on free quantum simulators during the POC phase, with a direct migration path to real NISQ hardware as the project matures.

Path from POC to Product

The POC validates the full pipeline on a cohort of 20 to 30 students across one or two subject modules, producing the behavioral dataset and convergence benchmarks needed to make the case for wider deployment. The step to an institutional pilot requires only an LTI 1.3 plugin, allowing NeuroAdapt to sit

inside an existing Moodle or Canvas installation without asking any institution to change its infrastructure. From there, the product operates as a B2B SaaS subscription priced per student annually, with a freemium tier for individual educators serving as the primary adoption channel.

Testing and Ablation Feasibility

The primary evaluation compares three matched cohorts: a static LMS control, a classical DDQN group, and the full quantum-classical hybrid. This three-arm structure isolates the quantum layer's contribution specifically, which prior work in this space has not cleanly done. Three ablation tests run in parallel: removing individual Observer flags to identify which signals are genuinely load-bearing, comparing the full six-action space against a reduced three-action version to evaluate whether the additional generative interventions justify their API cost, and directly comparing VQC convergence speed against an equivalent classical dense layer on the Preference Delta metric. All three tests are feasible within the Phase 3 timeline using the Phase 1 baseline dataset, with no additional data collection required.

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